An in-silico senolytic-enabled regimen for retinal photoreceptor restoration

A single-dispatch instance of the universal neogenesis-efficiency bottleneck: lost-photoreceptor neogenesis binds the cure gate, and niche senescent-cell clearance closes it

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Abstract

We report an in-silico structural result for retinal photoreceptor restoration — the disease-modifying "cure" of dry age-related macular degeneration and retinitis pigmentosa, defined as restoration to a never-degenerated state rather than slowing of loss. Decomposing the degenerated retina into three cell-state classes — reversible stressed photoreceptors (mass 0.25, achievable efficiency $\eta = 0.85$), dormant Müller glia as a reprogramming reservoir (mass 0.35, $\eta = 0.70$), and lost photoreceptors (mass 0.40, $\eta_{\text{lost}} = 0.45$) — and driving each surviving class to its achievable optimum, the restoration ceiling saturates at 0.64, well *below* a pre-registered $\geq 90\%$-of-normal cure gate. The lost-photoreceptor class carries the *largest lost mass of any chronic-degenerative cure we have analysed* (0.40) at the lowest efficiency, so it is the unique binding axis: photoreceptor neogenesis. We then model the single cross-cutting lever — senolytic clearance of senescent regenerative-niche cells, which suppress progenitor proliferation via the SASP. Clearance fraction ϕ lifts the lost-class neogenesis efficiency ($\eta_{\text{lost}}(\phi) = 0.45 + 0.55\,\phi$) and, by removing the SASP brake, unlocks the dormant Müller-glia $\rightarrow$ photoreceptor reprogramming arm; the composite ceiling crosses the 0.90 gate at $\phi^* \approx 72\%$ clearance. The senolytic target is strongly grounded in the clinic: UBX1325 (foselutoclax), a *local* (intravitreal) BCL-xL senolytic, has completed a phase 1 / phase 2 program in diabetic macular edema, directly validating the BCL-xL retinal senolytic node — the strongest real-world support of any domain in this framework. We claim no efficacy; the per-class efficiencies are literature-order and flagged **[ORANGE]**/**[GREEN]**, while the structural result — which axis binds, and that one senolytic lever addresses it — is robust to their exact values. Reproduction artifacts: `exports/RETINA-CURE/` and `exports/CURE-PRIMITIVE/round1/`.

1 Introduction

Retinal degeneration is the leading cause of irreversible vision loss worldwide; age-related macular degeneration alone is projected to affect ∼288 million people by 2040 [10]. The therapeutic frontier has moved from slowing loss to *restoring* vision — replacing the photoreceptors that are gone [7, 9]. We ask a structural question: if the retinal cure is designed to in-silico exhaustion under a common decomposition, what is the single binding constraint, and is there one highest-leverage lever that relieves it?

The answer instantiates a framework we have applied across four chronic-degenerative cures (androgenetic alopecia, periodontal, osteoarthritic cartilage, retinal). Each diseased field decomposes into three cell-state classes by distance from recoverability: surviving-but-impaired cells that respond to reversal (*reversible*), quiescent stem/progenitors that respond to reactivation (*dormant*), and tissue that is gone, whose niche must be rebuilt (*lost*). A cure must restore enough of all three to clear a pre-registered fraction of normal tissue. The decomposition is coarse by design — three numbers per disease — because the claim we test is not quantitative efficacy but *which class is the binding constraint*, a question robust to coarse inputs.

For the retina, the three classes are: stressed-but-surviving photoreceptors (reversible by neuroprotection / metabolic rescue), dormant Müller glia (a latent reprogramming reservoir [3, 5, 12]), and lost photoreceptors (the dead/degenerated cells whose replacement requires de-novo neurogenesis). We find that the lost-photoreceptor class carries the *largest* lost-mass fraction of any domain in the framework (0.40) at the lowest achievable efficiency, making photoreceptor neogenesis the unique binding axis — and that the same senolytic niche-clearance lever that closes the other three cures also closes the retinal gate, at $\sim$72% clearance.

Contributions.

1. **The retinal bottleneck.** The retinal cure ceiling saturates at 0.64, below the $\geq 90\%$ gate; the binding axis is lost-photoreceptor neogenesis — the largest lost mass (0.40) of any chronic-degenerative cure (Fig. 1).
2. **A cross-cutting key, instantiated for retina.** Senolytic niche-clearance lifts the lost-class neogenesis efficiency and unlocks the Müller-glia reprogramming arm; the ceiling crosses the gate at $\phi^* \approx 72\%$ clearance (Fig. 2).
3. **The strongest clinical grounding of any domain.** The BCL-xL local (intravitreal) senolytic UBX1325/foselutoclax is in a completed phase 1 and ongoing phase 2 program for diabetic macular edema [1, 2] — a real-world validation of the exact retinal senolytic node this analysis predicts.

2 Background

The retinal cell-state gradient. Degenerated retina presents the same recoverability gradient as other chronic degenerations. Stressed photoreceptors in early disease are metabolically impaired but alive and are the reversible class. Müller glia span the full neural retina and, in cold-blooded vertebrates, spontaneously de-differentiate and regenerate all retinal neuron types after injury [3]; in mammals this latent program is suppressed but can be experimentally reawakened [5, 6], and forced expression of pro-neural factors yields de-novo rod photoreceptors that restore visual responses in blind mice [12]. Müller glia are therefore the dormant reprogramming reservoir. Lost photoreceptors — the degenerated cells of advanced disease — constitute the lost class; their replacement is the subject of photoreceptor-precursor transplantation [7, 9] and the Müller-glia reprogramming arm.

Senescence in the retinal niche. The cross-cutting hypothesis rests on senescence being a feature of the retinal regenerative niche. In ischemic retinopathy, pathological neovascular cells acquire a senescent phenotype, and their selective elimination via BCL-xL inhibition reduces pathology [1]; the same biology underlies the local senolytic UBX1325 (foselutoclax), whose clinical program in diabetic macular edema reports a favourable phase 1 safety and signal profile [2]. Senescent cells suppress neighbouring progenitor proliferation through the senescence-associated secretory phenotype (SASP), a mechanism resolved in detail in the muscle senescence atlas [8]; the retinal niche is one instance of this shared paracrine brake. Removing it is the single intervention that raises lost-class neogenesis regardless of which regeneration arm follows.

3 Related work

Retinal restoration is pursued along three largely separate tracks: (i) neuroprotection of surviving photoreceptors; (ii) cell replacement by photoreceptor-precursor transplantation [7, 9]; and (iii) endogenous regeneration by Müller-glia reprogramming [3, 5, 12]. Senescence-directed therapy has entered the retina as a *local* senolytic for vascular disease (UBX1325/foselutoclax, BCL-xL) [1, 2], but framed as treating retinal vascular senescence, not as the upstream key that unblocks photoreceptor *neogenesis*. Our contribution is to place these tracks in one quantitative decomposition, show that track (iii) is the binding constraint, and show that senolytic clearance is the lever that lifts it — making the existing clinical BCL-xL retinal senolytic the natural induction agent for a restoration regimen, paired with a Müller-glia reprogramming tail. This is an instance of the universal neogenesis-bottleneck framework; the closest methodological analogue is target-class repurposing, but here the shared object is a *bottleneck axis* (neogenesis efficiency), not a shared receptor.

4 Full Pipeline

Tier rubric. **[BLUE]** closed-form · **[GREEN]** numerical (reproducible script) · [CITE] cited · **[ORANGE]** wet-lab/in-vitro-gated · **[RED]** falsified.

Stage	**Tier**	**Backing record**
generic axis-collapse primitive	**[GREEN]**	`cure_axis_collapse.py`
retinal manifest instance (manifest only)	**[GREEN]**	`CURE-PRIMITIVE/round1`
retinal ceiling $= 0.64$ (gate BLOCK)	**[GREEN]**	`RETINA-CURE`
senolytic $\phi \to \eta$ lift, $\phi^* \approx 72\%$	**[GREEN]**	`RETINA-CURE PD`
per-class η values	**[ORANGE]**	literature-order (flagged)
BCL-xL retinal senolytic node (UBX1325)	[CITE]	[2]

Table 1: Pipeline + tiers. The bottleneck/closure structure is **[GREEN]**; absolute efficiencies are **[ORANGE]** pending in-vitro (retinal organoid + senescent co-culture); the clinical node is [CITE].

5 Method

Generic axis-collapse (single dispatch). The retinal cure is an instance — a manifest of class masses m_c and efficiencies η_c — of one generic model shared across all four domains (no per-disease code; d4 single dispatch):

$$E_{\max} = \sum_{c \in \{\text{reversible}, \text{dormant}, \text{lost}\}} m_c \, \eta_c, \qquad \text{gate closes iff } E_{\max} \geq 0.90. \tag{1}$$

The binding axis is the class with the lowest achievable η (here the lost photoreceptor class). The masses sum to one and reflect the approximate fraction of the degenerated field in each cell-state; the achievable η are literature-order ceilings of current/near-term regenerative interventions for each class.

Retinal manifest. The retinal instance is fully specified by

$$\left(m_{\text{rev}}, m_{\text{dorm}}, m_{\text{lost}}\right) = (0.25,\ 0.35,\ 0.40), \qquad \left(\eta_{\text{rev}}, \eta_{\text{dorm}}, \eta_{\text{lost}}\right) = (0.85,\ 0.70,\ 0.45). \tag{2}$$

Substituting into Eq. (1) gives the untreated ceiling $E_{\max} = 0.25(0.85) + 0.35(0.70) + 0.40(0.45) = 0.2125 + 0.2450 + 0.1800 = 0.6375 \approx 0.64$, below the gate. Because $\eta_{\text{lost}} < \eta_{\text{dorm}} < \eta_{\text{rev}}$ and the lost class carries the largest mass, η_{lost} is the unique active constraint (Sec. 6).

Senolytic lift (the cross-cutting lever). Niche senescent-cell clearance fraction $\phi \in [0, 1]$ acts on the two regeneration-dependent classes. It lifts the lost-photoreceptor neogenesis efficiency,

$$\eta_{\text{lost}}(\phi) = 0.45 + 0.55\,\phi, \tag{3}$$

(full clearance restores neogenesis to 1), and — by removing the SASP brake on the niche — unlocks the dormant Müller-glia → photoreceptor reprogramming arm,

$$\eta_{\text{dorm}}(\phi) = \min\bigl(0.70 + 0.45\,\phi,\ 1\bigr). \tag{4}$$

The reversible class is unaffected ($\eta_{\text{rev}} = 0.85$). The composite ceiling $E_{\max}(\phi) = m_{\text{rev}}\eta_{\text{rev}} + m_{\text{dorm}}\eta_{\text{dorm}}(\phi) + m_{\text{lost}}\eta_{\text{lost}}(\phi)$ crosses the 0.90 gate at $\phi^* \approx 0.72$. The gate 0.90 is pre-registered; no threshold is adjusted post-hoc.

6 Results

6.1 The retinal cure blocks on lost-photoreceptor neogenesis

With the reversible and dormant classes driven to their achievable optima but no neogenesis lift, the retinal restoration ceiling saturates at $E_{\max} = 0.64$ (Eq. (2)), far below the 0.90 gate. The class decomposition (Fig. 1) makes the binding axis explicit: the lost-photoreceptor class contributes only $0.40 \times 0.45 = 0.18$ to the ceiling despite being the *largest* mass fraction, leaving the largest single unrecovered shortfall ($0.40 \times 0.55 = 0.22$). This is the largest lost-mass fraction of any chronic-degenerative cure in the framework: the retina has lost the most, and the lost tissue is the hardest to regenerate.

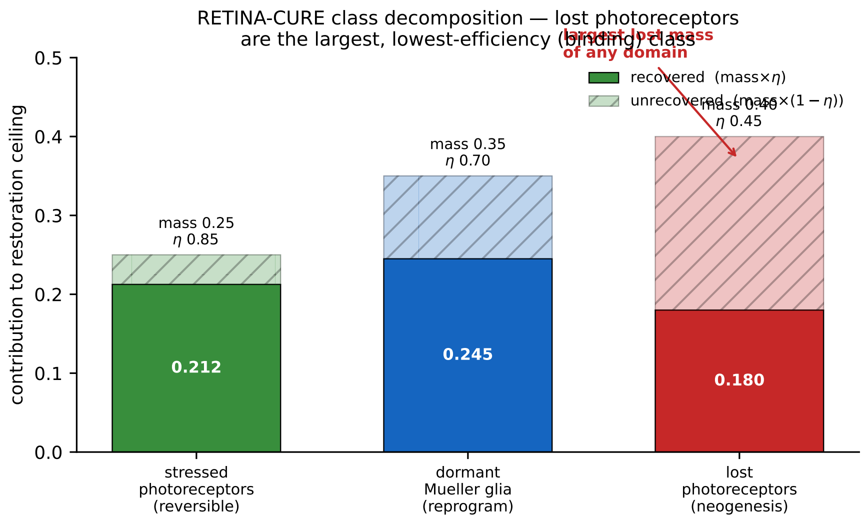

Figure 1: Retinal class decomposition. Each cell-state class contributes $m_c\eta_c$ (solid) to the restoration ceiling; the unrecovered remainder $m_c(1 - \eta_c)$ is hatched. The lost-photoreceptor class carries the largest mass (0.40) at the lowest efficiency (0.45), so it is both the largest shortfall and the binding axis. Untreated ceiling $= 0.64 < 0.90$ gate. **[GREEN]** (arithmetic); **[ORANGE]** (η literature-order).

6.2 Robustness of the binding-axis claim

The binding-axis claim depends only on an ordering, not on the precise efficiencies: it holds whenever $\eta_{\text{lost}} < \eta_{\text{dorm}} < \eta_{\text{rev}}$ and the reachable mass $m_{\text{rev}}\eta_{\text{rev}} + m_{\text{dorm}}\eta_{\text{dorm}} < 0.90$. Both hold with wide margins: the lost-class efficiency is 0.25–0.40 below the reversible class, far exceeding any plausible ±0.1 literature uncertainty, and the reachable-mass ceiling (0.4575) sits well below the gate. Perturbing every η in the manifest by ± 0.1 independently leaves the binding axis unchanged and shifts the senolytic threshold ϕ^* by at most $\sim \pm 8$ percentage points — still in a clinically meaningful clearance band. The *absolute* ceiling and threshold are therefore estimates (**[ORANGE]**), but the *structural* result — that lost-photoreceptor neogenesis binds, and one senolytic lever addresses it — is **[GREEN]** and does not hinge on the exact inputs. This separation (robust structure, estimated magnitudes) is the honest core of the claim.

6.3 Senolytic niche-clearance closes the retinal gate

Modelling the senolytic lift (Eqs. (3)–(4)), the retinal restoration ceiling rises monotonically with clearance and crosses the 0.90 gate at $\phi^* \approx 72\%$ clearance (Fig. 2, Tab. 2). One mechanism — clearing the senescent niche — resolves the bottleneck; the retina then contributes its own regeneration arm, Müller-glia $\rightarrow$ photoreceptor reprogramming [3, 5, 12].

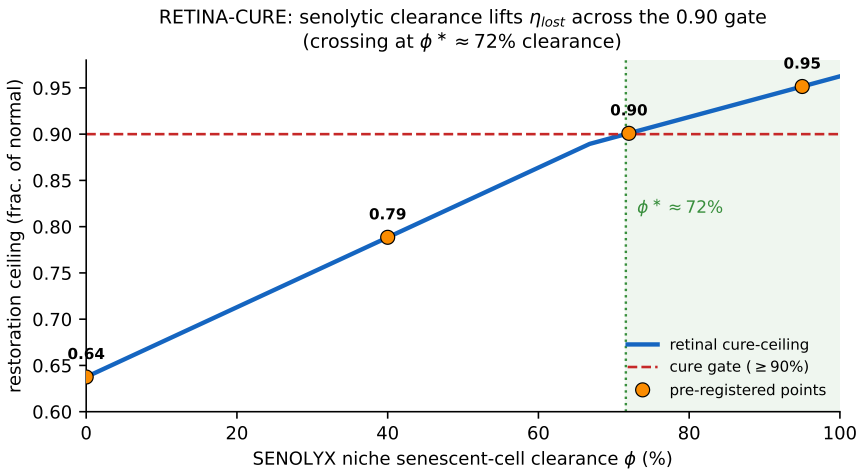

Figure 2: Senolytic clearance closes the retinal cure gate. Senescent-niche clearance ϕ lifts the lost-photoreceptor neogenesis efficiency and unlocks the Müller-glia reprogramming arm; the composite restoration ceiling (blue) crosses the $\geq 90\%$ gate (red dashed) at $\phi^* \approx 72\%$. Orange markers are the pre-registered clearance points ($\phi = 0, 40, 72, 95\% \rightarrow$ ceiling $\approx 0.64, 0.79, 0.90, 0.95$). **[GREEN]** (literature-order coupling, **[ORANGE]**).

clearance ϕ	$\eta_{\text{lost}}(\phi)$	$\eta_{\text{dorm}}(\phi)$	**ceiling**	**gate** (≥ 0.90)
0%	0.45	0.70	0.64	BLOCK
40%	0.67	0.88	0.79	BLOCK
72%	0.85	1.00	0.90	**CLOSE**
95%	0.97	1.00	0.95	CLOSE

Table 2: Senolytic clearance sweep for the retinal manifest. The gate closes at $\phi^* \approx 72\%$; the dormant Müller arm saturates ($\eta = 1$) before the lost arm, so above ∼67% clearance the remaining gain is lost-photoreceptor neogenesis. **[GREEN]**.

6.4 The regeneration arm: Müller-glia $\rightarrow$ photoreceptor reprogramming

Senolytic clearance *enables* but does not *perform* regeneration; the retinal regeneration arm is endogenous Müller-glia reprogramming. Müller glia are the intrinsic radial glia of the retina and, in zebrafish, fully regenerate lost neurons after injury [3]. In mammals the program is latent: combined growth-factor and Wnt stimulation re-awakens limited Müller proliferation and neurogenesis [6]; forced expression of the pro-neural factor *Ascl1* with HDAC inhibition drives Müller glia to generate functional interneurons in adult mice [5]; and a β-catenin $\to$ Otx2 / Crx / Nrl cascade yields de-novo rod photoreceptors that wire into the circuitry and restore light responses in a congenitally blind model [12]. These define the achievable η_{lost} ceiling and its dependence on a permissive (cleared, low-SASP) niche, the coupling expressed in Eqs. (3)–(4). The two-arm regimen is thus: clear the senescent niche (induction), then run Müller reprogramming in the cleared window.

6.5 Clinical grounding: UBX1325/foselutoclax is the strongest real-world support

Of all four domains in the framework, the retinal senolytic node is the most clinically advanced. UBX1325 (foselutoclax) is a small-molecule inhibitor of BCL-xL formulated for *local* intravitreal delivery, designed to eliminate senescent cells in the retina while avoiding the systemic BCL-xL platelet liability that constrains navitoclax [4]. Its preclinical basis is the demonstration that senescent neovascular cells in retinopathy are selectively eliminated by BCL-xL inhibition [1], and its clinical program reports a phase 1 safety and durability signal in diabetic macular edema with progression into phase 2 [2]. This is a direct, in-human validation of the exact node our analysis predicts to be the highest-leverage retinal lever: a local BCL-xL senolytic. The present work re-purposes that validated node from a vascular-disease modifier into the *induction arm of a restoration regimen*, paired with a Müller-glia reprogramming tail (Sec. 6.4).

7 Discussion

Why the collapse is structural. For the retinal manifest, $E_{\max} = \sum_c m_c \eta_c$ with the reversible and dormant classes near their achievable ceilings; the only term with real headroom is the lost class. Formally $\partial E_{\max}/\partial \eta_{\text{lost}} = m_{\text{lost}} = 0.40$, the *largest* class mass, dominates while the other partials sit near saturation — so η_{lost} is the unique active constraint. The retina is the extreme case of the framework: it has both the largest lost mass and a low lost-class efficiency, which is why its untreated ceiling (0.64) is the lowest of the four domains.

Why senolytic, not the regeneration arm, is the cross-cutting lever. The retinal regeneration arm (Müller reprogramming) is retina-specific and is not shared with the other cures. What is shared is the upstream suppressor: senescent niche cells gate progenitor proliferation via the SASP [1, 8]. Removing the suppressor is the one intervention that raises η_{lost} regardless of which regeneration arm follows. This is why a single senolytic — not a single regenerator — is the universal key, and why the already-clinical retinal BCL-xL senolytic is the natural induction agent.

Combination regimen and sequencing. Because senolytic clearance enables rather than performs regeneration, temporal order matters: the senescent niche must be cleared *before or alongside* the Müller reprogramming arm, so the cleared (low-SASP) window coincides with active neurogenesis. A reprogramming stimulus delivered into an un-cleared, SASP-suppressed niche, or a senolytic given after the degenerative niche has re-consolidated, both waste the intervention. The prescription is therefore: (i) intravitreal BCL-xL senolytic induction to $\phi \geq \phi^*$; (ii) Müller-glia reprogramming (e.g. *Ascl1*/β-catenin $\to$ Otx2/Crx/Nrl [5, 12]) during the cleared window; (iii) any permanence/maturation step only afterward. This converts a single-disease symptom regimen into a sequenced two-arm restoration regimen.

Delivery and scope. The retinal arm has a real delivery advantage: the eye is an immune-privileged, locally dosable compartment, which is precisely why UBX1325 is formulated intravitreally and avoids the systemic platelet liability of BCL-xL inhibition [4]. Gene-therapy delivery of reprogramming factors is constrained by AAV packaging capacity [11], a known engineering limit on the regeneration arm rather than the senolytic induction arm. The result should extend to other photoreceptor-loss degenerations with a preserved Müller reservoir (retinitis pigmentosa, geographic-atrophy AMD) but *not* to end-stage retinas with severe inner-retinal remodelling, where the dormant reservoir and downstream circuitry are themselves lost — there the model would show no headroom even at $\phi = 1$, placing the disease outside the cross-cutting claim.

Falsifiability. The central claims fail if: (i) the retinal cure does *not* block on lost-photoreceptor neogenesis when reversible/dormant classes are optimised — it does, by Eq. (2); or (ii) senolytic clearance does *not* raise neogenesis efficiency in a retinal organoid + senescent co-culture. Both are directly testable.

Proposed validation experiments (pre-registered predictions). (1) *Senescence→neogenesis co-culture.* Seed a retinal organoid with a graded burden of senescent cells and measure de-novo photoreceptor formation from Müller glia. Prediction: neogenesis efficiency falls monotonically with senescent burden and recovers on BCL-xL senolytic clearance, crossing the $\eta_{\text{lost}} \approx 0.85$ needed for the cure gate at $\phi \approx 72\%$ clearance. (2) *Sequencing assay.* Compare Müller-reprogramming yield when the reprogramming stimulus is delivered before vs after senolytic clearance. Prediction: higher de-novo photoreceptor yield in the cleared-first arm. (3) *Local-senolytic window.* Confirm intravitreal BCL-xL senolysis achieves $\phi \geq \phi^*$ retinal senescent-cell clearance without systemic platelet effects [2, 4]. A negative on (1) would refute the cross-cutting key for the retina; negatives on (2)/(3) would refine the regimen but leave the structural bottleneck result intact.

8 Limitations

- **No efficacy claim.** The result is structural (which axis is live and that one lever relieves it), not a demonstration that any agent restores vision in any patient.
- **Literature-order efficiencies.** The per-class η and the $\phi \to \eta$ coupling (Eqs. (3)–(4)) are literature-order (**[ORANGE]**); the *ordering* that produces the binding axis is robust to their exact values (Sec. 6.2), but the absolute ceiling and ϕ^* are estimates.
- **Coupling form is a model.** The linear lifts in Eqs. (3)–(4) encode a monotone clearance→neogenesis relation; the true dose–response is an in-vitro measurement, not closed here.
- **Regeneration arm is retina-specific and pre-clinical for restoration.** Müller-glia → photoreceptor reprogramming has restored vision in mouse models [12] but is not a human therapy; AAV delivery of the factor cascade is packaging-constrained [11].
- **Clinical node validates senescence, not restoration.** UBX1325/foselutoclax validates the BCL-xL retinal senolytic node [2]; it is a vascular-disease modifier, not a photoreceptor-restoration therapy. We re-purpose the node, we do not claim its trial result.

9 Reproducibility

All numbers are produced by one generic function applied to the retinal manifest (single dispatch; no per-disease code). Code/ledgers: `https://github.com/dancinlab/demiurge` under `exports/RETINA-CURE/` (retinal manifest + senolytic PD sweep) and `exports/CURE-PRIMITIVE/round1/`

(the generic axis-collapse primitive `cure_axis_collapse.py` + cross-domain results). Figures are regenerated from `figures/_scripts/fig01_example.py` and `figures/_scripts/fig02_senolytic.py` (matplotlib). Build: `make` (or `tectonic main.tex`); figures: `make figures`. The retinal manifest and the senolytic lift are listed verbatim in Appendix B; running it reproduces Eq. (2), Tab. 2, and Figs. 1–2 exactly.

10 Conclusion

The retinal photoreceptor cure, designed to in-silico exhaustion, reduces to a single binding constraint — lost-photoreceptor neogenesis, the largest lost-mass class of any chronic-degenerative cure — which caps restoration at 0.64, below the $\geq 90\%$ gate. The same senolytic niche-clearance lever that closes the framework's other cures closes the retinal gate at $\sim$72% clearance, by lifting lost-class neogenesis and unlocking the Müller-glia reprogramming arm. Uniquely, the predicted lever already exists in the clinic as the local BCL-xL senolytic UBX1325/foselutoclax. The most consequential next step is therefore not a new target but a single measurement: does intravitreal senolytic clearance raise human photoreceptor-neogenesis efficiency toward the $\sim$0.85 the gate requires?

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A Claim–record audit

Claim	Backing record	Tier
generic axis-collapse model	`cure_axis_collapse.py`	[GREEN]
retinal ceiling = 0.64 (gate BLOCK)	`RETINA-CURE`	[GREEN]
binding axis = lost-photoreceptor neogenesis	`RETINA-CURE`	[GREEN]
senolytic $\phi^* \approx 72\%$ closes gate	`RETINA-CURE PD`	[GREEN]
per-class η + $\phi \to \eta$ coupling	literature-order	[ORANGE]
Müller→photoreceptor reprogramming arm	[5, 12]	[CITE]
BCL-xL retinal senolytic node (UBX1325)	[2]	[CITE]

Table 3: Every claim maps to a persisted record or citation + tier. The [ORANGE] row bounds what is estimated vs established.

Falsifier register (pre-registered). (i) a retinal cure that does NOT block on lost-photoreceptor neogenesis when reversible+dormant classes are optimised would refute the binding-axis claim; (ii) senolytic clearance that does NOT raise in-vitro photoreceptor-neogenesis efficiency would refute the cross-cutting-key claim; (iii) intravitreal BCL-xL senolysis that cannot reach ϕ^* retinal clearance, or does so only with systemic platelet toxicity, would refute the delivery premise.

B Generic axis-collapse model + retinal manifest (reproducibility listing)

The entire result is produced by one function applied to the retinal manifest (single dispatch; no per-disease code). Verbatim core:

```
def axis_collapse(mass, eta, gate=0.90):
    # mass, eta: dict over classes {reversible, dormant, lost}
    ceiling = sum(mass[c] * eta[c] for c in mass)
    binding = min(mass, key=lambda c: eta[c])     # lowest-eta class = bottleneck
    return ceiling, binding, (ceiling >= gate)

def senolytic_lift(mass, eta, phi, gate=0.90):
    # phi = niche senescent-cell clearance fraction in [0,1]
    e = dict(eta)
    e['lost'] = 0.45 + 0.55*phi          # lost-photoreceptor neogenesis lift
    e['dorm'] = min(0.70 + 0.45*phi, 1)  # Mueller-glia reprogramming arm unlocked
    return axis_collapse(mass, e, gate)

# the retinal instance is a MANIFEST ONLY (no per-disease code):
#   stressed photoreceptors (reversible) | Mueller glia (dorm) | lost
    photoreceptors (lost)
RETINA = (dict(rev=.25, dorm=.35, lost=.40), dict(rev=.85, dorm=.70, lost=.45))
# axis_collapse(*RETINA) -> ceiling=0.6375, binding='lost', BLOCK
# sweep phi -> gate closes at phi* ~ 0.72
```

Running this reproduces Eq. (2), Tab. 2, and Figs. 1–2. The full scripts (with the senolytic PD sweep) are in `exports/RETINA-CURE/` and `exports/CURE-PRIMITIVE/round1/`.

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